

UD65640

Application Note

QSPI

AN-K0015-250807

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1. Pin Descriptions

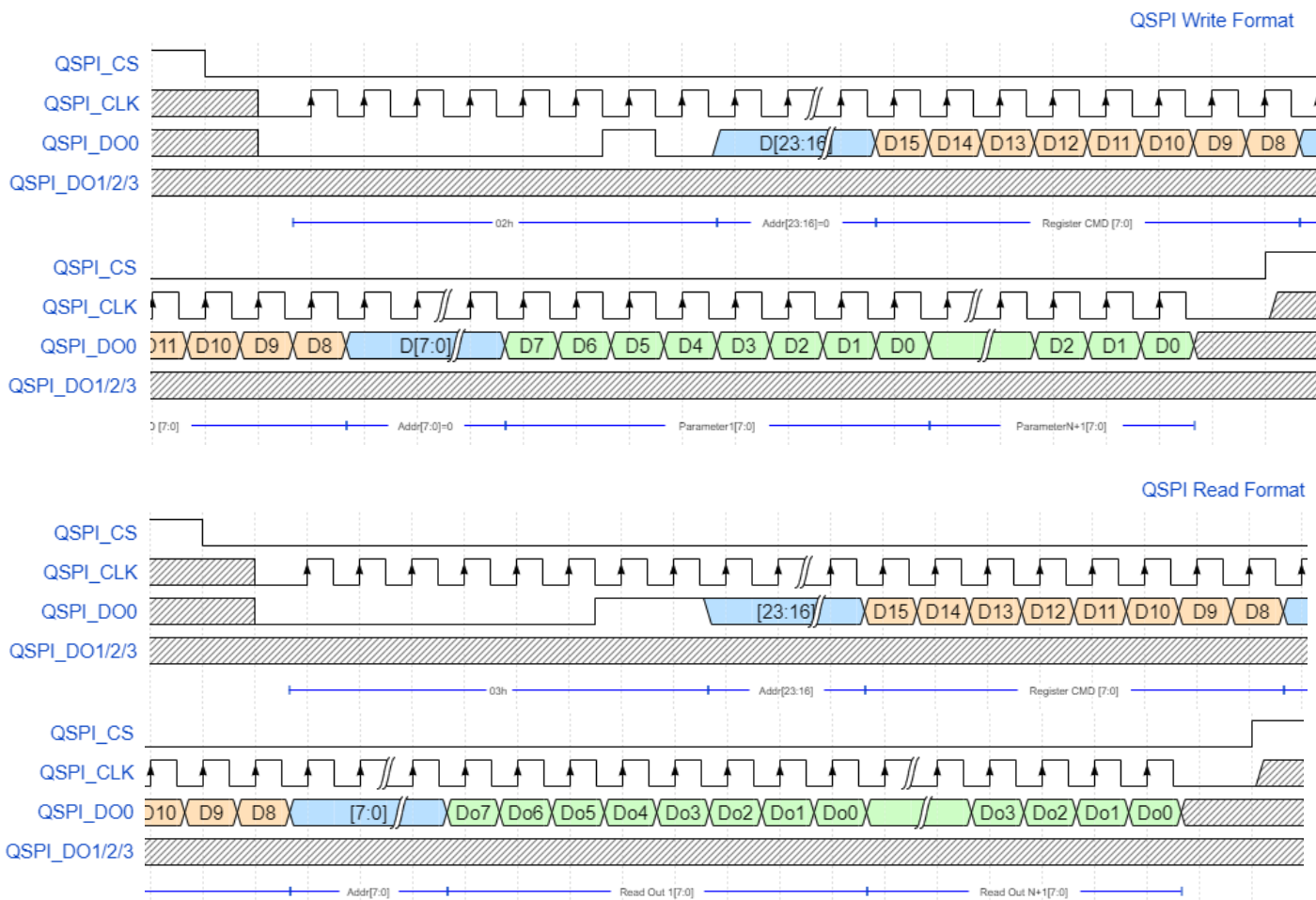
This description applies to the QSPI interface.

Pin No.	Symbol	Type	Pin Description	When not used
1	NTEST	Output	Analog test pin, leave open.	Open
2	VPP	Power Supply	OTP programming power, leave open when not used.	Open
3	ELVDD	Power Supply	Power supply for OLED cell, connect a capacitor for stabilization.	NA
4	ELVSS	Power Supply	Power supply for OLED cell, connect a capacitor for stabilization.	NA
5	IOVCC	Power Supply	1.8V or 3.3V Power supply for interface system.	NA
6	GPIO1	Output	Leave open.	Open
7	QSPI_CLK	Input	Serial clock for QSPI interface	NA
8	QSPI_CSX	Input	Chip select for QSPI interface	NA
9	QSPI_D0	Input/Output	Serial data0 input and output for QSPI interface	NA
10	QSPI_D1	Input	Serial data1 input for QSPI interface	NA
11	QSPI_D2	Input	Serial data2 input for QSPI interface	NA
12	QSPI_D3	Input	Serial data3 input for QSPI interface	NA
13	Reserved	Output	Leave open.	Open
14	Reserved	Output	Leave open.	Open
15	Reserved	Output	Leave open.	Open
16	Reserved	Output	Leave open.	Open
17	Reserved	Output	Leave open.	Open
18	Reserved	Output	Leave open.	Open
19	Reserved	Output	Leave open.	Open
20	Reserved	Output	Leave open.	Open
21	Reserved	Input	Leave open.	Open
22	BM1	Input	Connect to IOVCC	NA
23	TE	Output	Tearing effect output, signal to synchronize driver	NA
24	GPIO0	Output	Leave open.	Open

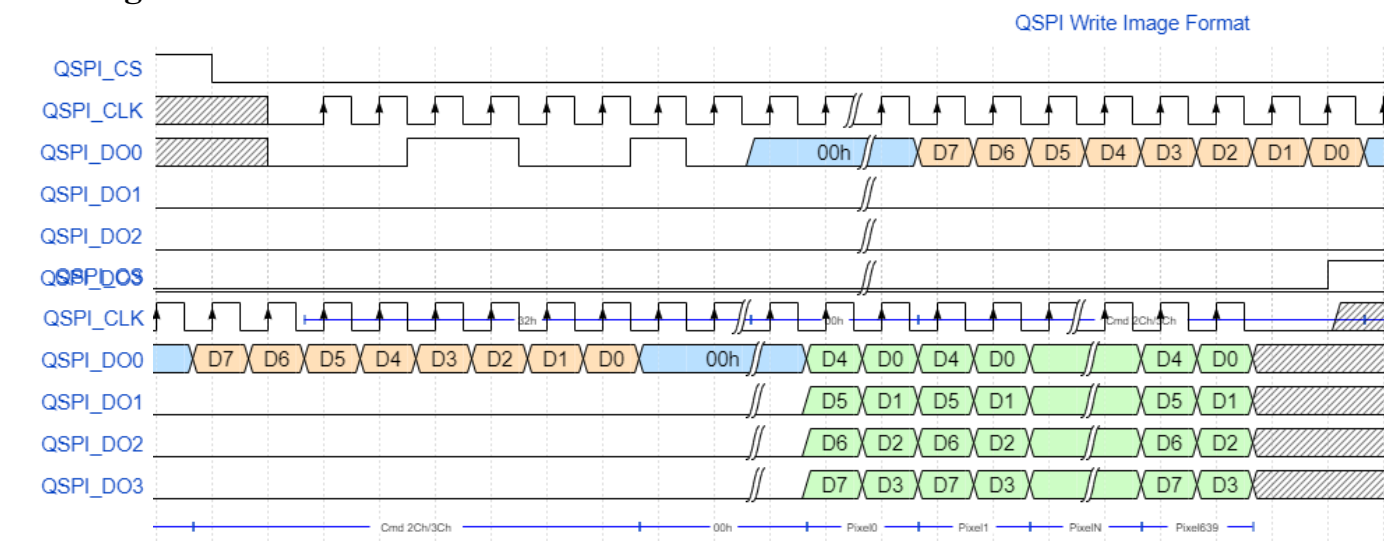
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25	Reserved	Output	Leave open.	NA
26	Reserved	Output	Leave open.	NA
27	Reserved	Output	Leave open.	NA
28	Reserved	Output	Leave open.	NA
29	Reserved	Output	Leave open.	NA
30	Reserved	Output	Leave open.	NA
31	Reserved	Output	Leave open.	NA
32	Reserved	Output	Leave open.	NA
33	Reserved	Output	Leave open.	NA
34	SWIRE	Output	SWIRE output pin used to control the power IC (ELVSS power). SWIRE Pulse can from Register Programmable	Open
35	TESTE	Input	Test pin, leave open.	Open
36	ATEST	Output	Analog test pin, leave open.	Open
37	AVCI33	Power Supply	3.3V Power supply for analog circuit, connect a capacitor for stabilization.	N/A
38	AVSS	Power Supply	Analog ground.	N/A
39	DVSS	Power Supply	Digital ground.	N/A
40	DVDD	Power Supply	Regulator output to generate voltage for internal logic, connect a capacitor for stabilization.	N/A
41	AVCI18	Power Supply	1.8V Power supply for analog circuit, connect a capacitor for stabilization.	N/A
42	RESX	Input	Active-low reset pin.	N/A
43	SDA0	Input	Pin used to set the I2C device address.	DVSS/IOVCC
44	SDA	Input/Output	Bi-directional data pin for the I2C interface. Connect to IOVCC if not used.	IOVCC
45	SCL	Input	Synchronous clock signal pin for the I2C interface. Connect to IOVCC if not used.	IOVCC
46	AVCI18	Power Supply	1.8V Power supply for analog circuit, connect a capacitor for stabilization.	N/A
47	ELVSS	Power Supply	Power supply for OLED cell, connect a capacitor for stabilization.	N/A
48	ELVDD	Power Supply	Power supply for OLED cell, connect a capacitor for stabilization.	N/A
49	BM0	Input	Connect to DVSS	DVSS
50	Reserved	Input	Leave open.	Open

2. Command Format

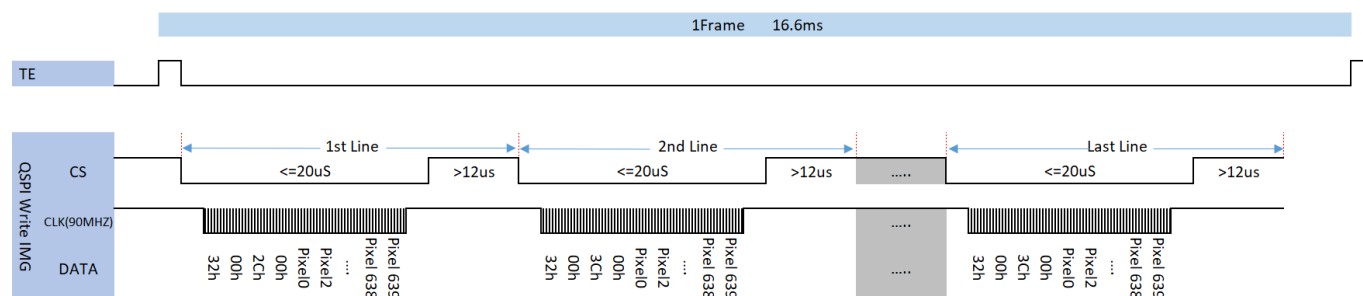


3. Image Format



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4. On TE Timing



5. Register

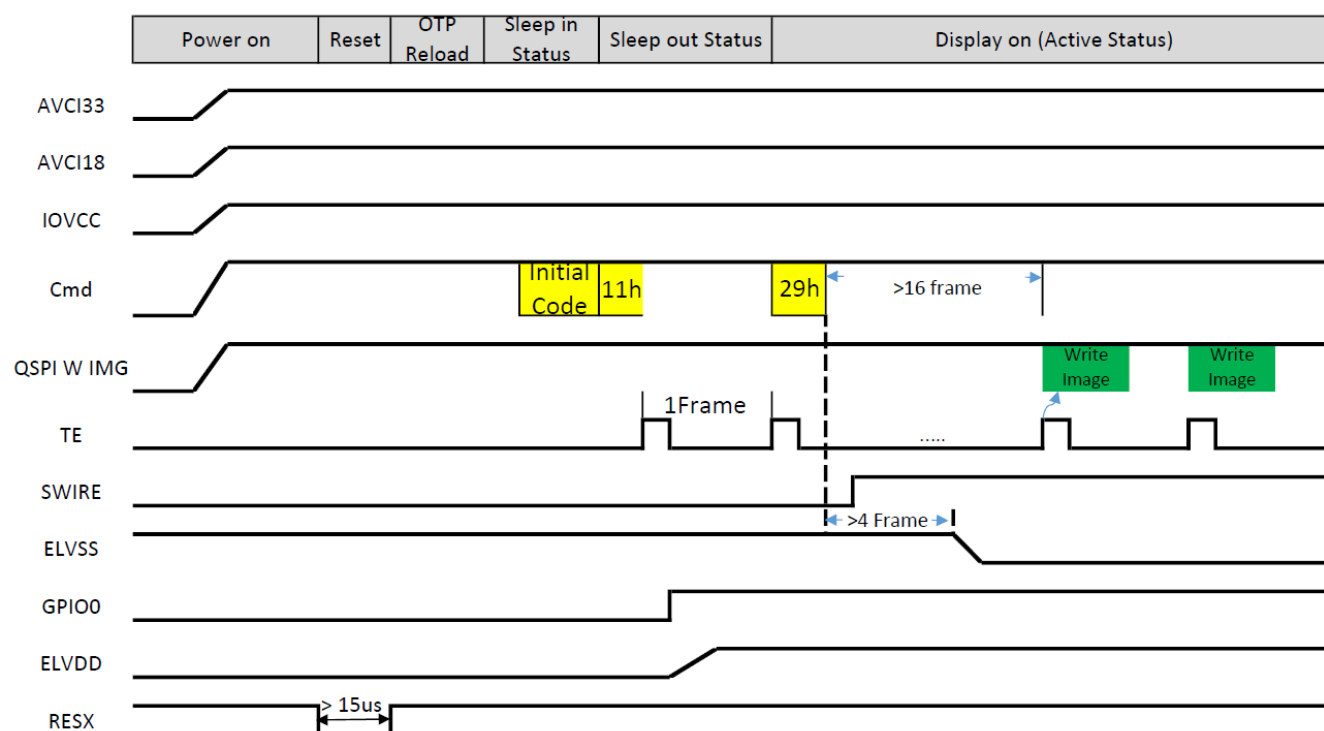
MCS Ctrl: Manufacturer Command Set Control (80h)

80h	MCS Ctrl									
	W / R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	0	0	0	0	80h
parameter 1	W / R	0	0	0	0	0	0	0	MCS_EN	00h
Descriptions	MCS_EN This command releases parameters so that Manufacturer Command inputs are enabled. 1: Access enabled. 0: Access denied. (Protection on) 1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restrictions										

Interface Control: (8Dh)

8Dh	Interface Control									
	W / R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	1	1	0	1	8Dh
parameter 1	W / R	1	0	DM	0	0	0	0	0	00h
Descriptions	DM : 0 : Command Mode (MIPI Mode or QSPI Mode) 1 : Video Mode 1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restrictions										

6. Power Sequence



- Note:**
1. ELVSS power on timing is depend on power IC, we suggest customer to select power IC that power on timing is > 4 frame.
 2. ELVDD can adjust the opening time through the register.
 3. Write Image triggers at least 16 frame to send the image.



7. Initial Code For 60Hz (Version UD65640B)

80,01

90,74,44,0A

89,84,84,00,09

8A,18,03,07,04

96,31,00

97,01,00

95,81,7F

A8,01

AA,15,04,00,24,00,00

A9,01

51,FF,03

35,00

44,01

11,00

29,00

Revision History

Version No.	Date	Page	Description
AN-K0004-240709	2024/07/09		Initial Version
AN-K0004-240730	2024/07/30	Page6	Max Brightness Current 95,02,81,7F Modify 95,81,7F
AN-K0015-241017	2024/09/10	Page 3	Pin Description
AN-K0015-241017	2024/09/22	Page 9	Add Power Sequence Delay Description
AN-K0015-250807	2024/08/07	Page 10	Power On Sequence add elvss / send image trigger TE times..